According to Charles's law,

$$V/T = constant \quad (3.2)$$

According to Avogadro's law,

$$V/n = constant \quad (3.3)$$

After combining equations (3.1), (3.2) and (3.3), you can get the following equation.

$$PV/nT = constant \quad (3.4)$$

The above relation is called the combined law of gases. If you consider a gas, which contains μ moles of the gas, the number of atoms contained will be equal to μ times the Avogadro number, N_A.

$$i.e. \quad n = \mu N_A. \quad (3.5)$$

Using equation (3.5), equation (3.4) can be written as

$$PV/ \mu N_A T = constant$$

The value of the constant in the above equation is taken to be k_B, which is called as **Boltzmann constant (1.38×10^{-23} JK^{-1}).** Hence, we have the following equation:

$$PV/ \mu N_A T = k_B$$
$$PV = \mu N_A k_B T$$

Here, $\mu N_A k_B = R$, which is termed as universal gas constant whose value is

$8.31\ J\ mol^{-1} K^{-1}$.

$$PV = RT \quad (3.6)$$

Ideal gas equation is also called as *equation of state* because it gives the relation between the state variables and it is used to describe the state of any gas.

Points to Remember

- The SI unit of heat energy absorbed or evolved is joule (J)
- Heat always flows from a system at higher temperature to a system at lower temperature.
- **Temperature** is defined as the degree of hotness of a body. The SI unit of temperature is kelvin (K).
- All the substances will undergo one or more of the following changes when heated:
  i) Temperature of the substance rises.
  ii) The substance may change state from solid to liquid or gas.
  iii) The substance will expand when heated.
- All forms of matter (solid, liquid and gas) undergo expansion on heating.
- For a given rise in temperature, a liquid will have more expansion than a solid and a gaseous substance has the highest expansion than the other two.
- If a liquid is heated directly without using any container, then the expansion that you observe is termed as **real expansion** of the liquid.
- The expansion of a liquid apparently observed without considering the expansion of the container is called the **apparent expansion** of liquid.
- For a given heat energy, the real expansion is always more than that of apparent expansion.
- If the atoms or molecules of a gas do not interact with each other, then the gas is said to be an **ideal gas** or a **perfect gas**.
- Ideal gas equation, also called as equation of state is PV = RT. Here, R is known as universal gas constant whose value is 8.31 J mol^{-1}K^{-1}

Solved Problems

Example 1

A container whose capacity is 70 ml is filled with a liquid up to 50 ml. Then, the liquid in the container is heated. Initially, the level of the liquid falls from 50 ml to 48.5 ml. Then we heat more, the level of the liquid rises to 51.2 ml. Find the apparent and real expansion.

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